

Alzikrayat Project Report

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Project Name: Alzikrayat Photo Sharing Application

Project Overview

Alzikrayat is a responsive web application for sharing and preserving personal photo memories. A visitor can explore the home page, gallery, and About Us page. A registered user can create an account, sign in, upload photos, view photo details, delete owned photos, and add comments.

The project was developed for the Advanced Web Technologies course. It demonstrates a manual Model–View–Controller architecture and a three-tier separation without using an external web framework or an object-relational mapper. The implementation is designed to run locally through XAMPP, Apache, PHP, and MySQL.

Architecture & Project Structure

The application uses `public/index.php` as a single front controller. It registers HTTP methods and URL patterns with a custom router in `core/Router.php`. The router converts patterns into regular expressions and captures parameters such as the numeric identifier in `/photo/25`. It then dispatches the captured value to the appropriate controller.

The presentation tier is implemented in the `views` directory. These templates use HTML5, Bootstrap 5, Bootstrap Icons, CSS, and JavaScript. The application tier contains controllers and shared helpers. `AuthController` manages registration and authentication, `PhotoController` manages photo operations, and `CommentController` manages comments. The data tier contains the PDO connection and the `User`, `Photo`, and `Comment` models. Each model uses prepared SQL statements to communicate with MySQL.

The main directories are shown below:

Directory	Responsibility
config	PDO database configuration
core	Router, bootstrap process, security and view helpers
controllers	Authentication, photo, and comment request handling
models	Database queries and entity operations
views	Layouts, authentication pages, gallery pages, and static pages
public	Front controller, CSS, JavaScript, and uploaded images

Database Design

The database is normalized into three related tables. The `users` table stores account and profile information. The `photos` table stores uploaded file metadata and links each photo to its owner. The `comments` table links written comments to both a photo and the user who wrote them.

Each table uses an auto-incrementing integer primary key. Foreign keys enforce referential integrity. Deleting a user cascades to that user's photos and comments. Deleting a photo cascades to its comments. Indexes support user lookups, gallery ordering, and comment retrieval.

Table	Main fields	Relationship
users	id , names, email, password, profile fields	One user owns many photos and comments
photos	id , user_id , file name, title, description, timestamp	Each photo belongs to one user
comments	id , photo_id , user_id , comment, timestamp	Each comment belongs to one photo and one user

Passwords are never stored as plain text. The application uses `bcrypt` through PHP's `password_hash` and checks credentials with `password_verify` . All SQL values supplied by users are bound through PDO placeholders.

Main Features

The application provides secure user registration with validation for names, email addresses, and password length. It supports session-based authentication and changes the navigation bar according to the current authentication state. Logged-in users see their first name, an Upload link, and a Logout button. Visitors see a Login / Register button.

After a successful login, the application stores the exact login timestamp in an HttpOnly cookie with a seven-day expiration. The login page can display the previous login timestamp from the same browser. Failed login attempts return to a clean login form with a clear error message instead of retaining the submitted password.

Authenticated users can upload JPG, PNG, GIF, and WEBP images under five megabytes. The application checks the upload error code, MIME type, file size, destination directory, and write permission. It generates a random server-side filename and stores the title, description, owner, and timestamp in MySQL. Users can delete only their own photos after an ownership check.

The gallery supports a responsive three-column layout and a list-card layout. The detail page displays the high-resolution image, title, author, description, timestamp, comments, and the authorized delete action. Logged-in users can add comments, which are stored with their user ID and the current timestamp.

The interface uses Bootstrap responsive components, a custom visual theme, consistent navigation, responsive cards, image previews, form validation, and accessible show-password controls on the login and registration pages.

Novelty/Advanced Features

The project includes a manual regular-expression routing engine rather than framework routing. This router captures dynamic URL parameters and dispatches them to controller methods while keeping routing logic visible for academic review.

The gallery provides multiple presentation modes so that users can choose between a three-column visual grid and a list-card layout. The application also includes a seven-day browser-specific last-login memory, ownership-aware deletion, MIME-based upload filtering, and a show-password control implemented with JavaScript and Bootstrap Icons.

The interface is responsive across desktop, tablet, and mobile screen sizes. The design uses a consistent visual identity with a dark navy base, warm gold accents, card-based content, hover states, and readable empty states for galleries without photos.

Conclusion

Alzikrayat satisfies the central requirements of the project specification. It demonstrates a custom MVC implementation, manual regular-expression routing, a normalized relational database, secure authentication, photo upload and deletion, comments, multi-layer validation, XSS and SQL injection defenses, and a responsive Bootstrap interface.

The application can be demonstrated locally through XAMPP using MySQL port `3306`. The supplied README provides the setup sequence, and the `schema.sql` file provides the complete database structure.

References